This can be rewritten using a single function header in the following manner

float area (float radius);

float area (float half, float base, float height);

float area (float length , float breadth);

Illustration 15.1 C++ Program to demonstrate function overloading

```
#include <iostream>
using namespace std;
void print(int i)
 {cout<< " It is integer " << i <<endl;}
void print(double  f)
{  cout<< " It is float " << f <<endl;}
void print(string c)
{  cout<< " It is string " << c <<endl;}
int main() {
        print(10);
        print(10.10);
        print("Ten");
        return 0;
}
```

Output:

```
It is integer 10
 It is float 10.1
It is string Ten
```

Tip Notes

Function overloading is not only implementing polymorphism but also reduces the number of comparisons in a program and makes the program to execute faster. It also helps the programmer by reducing the number of function names to be remembered.

15.2.2 Rules for function overloading

1. The overloaded function must differ in the number of its arguments or data types
2. The return type of overloaded functions are not considered for overloading same data type
3. The default arguments of overloaded functions are not considered as part of the parameter list in function overloading.

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